

A monotone average–max sensitivity construction with a fixed exponent above 2/3

Slava Naprienko

Abstract

The Servedio–Tan open problem asks whether every monotone Boolean function f satisfies $\text{AS}(f) = o(\text{S}(f))$ [FHH⁺14, ScTa13]. We construct monotone Boolean functions F_m with $\text{S}(F_m) \rightarrow \infty$ and

$$\text{AS}(F_m) \geq \text{S}(F_m)^{2/3+\eta}$$

for all m , where $\eta > 8 \cdot 10^{-4}$ is an absolute constant. This improves the best previous explicit average–max sensitivity tradeoff for monotone Boolean functions, due to O’Donnell and Servedio [OS07], who achieved exponent $\alpha_{\text{OS}} = \log_2(6 - 2\sqrt{5}) \approx 0.6115$.

1 Introduction

Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be a monotone Boolean function. Two standard complexity measures of f are its average sensitivity (also called total influence) $\text{AS}(f) = \text{Inf}_{1/2}(f)$, the average over a uniform input of the number of pivotal coordinates, and its (maximum) sensitivity $\text{S}(f)$, the corresponding worst-case quantity. Trivially $\text{AS}(f) \leq \text{S}(f)$, and Servedio and Tan (recorded under the heading *Average versus Max Sensitivity for Monotone Functions* in [FHH⁺14]) asked whether monotone functions in fact satisfy

$$\text{AS}(f) = o(\text{S}(f)) \quad \text{as } \text{S}(f) \rightarrow \infty.$$

In the more explicit form stated by Scheder and Tan [ScTa13], the conjecture predicts absolute constants $K > 0$ and $\delta < 1$ such that $\text{AS}(f) \leq K \cdot \text{S}(f)^\delta$ for every monotone f . A natural way to track progress is through the exponent

$$\alpha(f_n) = \liminf_{n \rightarrow \infty} \frac{\log \text{AS}(f_n)}{\log \text{S}(f_n)}.$$

The Servedio–Tan question asks whether $\alpha(f_n) < 1$ for every monotone sequence with $\text{S}(f_n) \rightarrow \infty$. In the opposite direction, O’Donnell and Servedio [OS07] gave a recursive construction achieving

$$\text{AS}(f_n) \geq \text{S}(f_n)^{\alpha_{\text{OS}} - o(1)}, \quad \alpha_{\text{OS}} = \log_2(6 - 2\sqrt{5}) \approx 0.6115,$$

by composing a four-variable base gadget. As noted in [ScTa13], this remained the best known explicit lower bound on α for monotone functions.

Remark 1. A small improvement over the O’Donnell–Servedio exponent is already obtained from the Fano-plane line-containment function. This monotone function has sensitivity 3 and average sensitivity $63/32$, so recursive composition gives exponent

$$\frac{\log(63/32)}{\log 3} \approx 0.6166,$$

slightly larger than $\alpha_{\text{OS}} \approx 0.6115$. We do not use this finite gadget below; the main construction gives a qualitatively stronger exponent above $2/3$.

Our main result is the following.

Theorem 2. *There exist an absolute constant $\eta > 0$ and a sequence of monotone Boolean functions F_m with $S(F_m) \rightarrow \infty$ such that*

$$\text{AS}(F_m) \geq S(F_m)^{2/3+\eta}$$

for every $m \geq 1$.

The geometric heart of the argument is an irredundant affine-hyperplane cover bound for $\mathbb{F}_Q^d$: it forces a $\log Q$ gap between the two one-sided sensitivities of an algebraic DNF block whose terms are graphs of low-degree polynomials. Pairing such a block with the dual of a second one yields a *biased cascade* whose output is again low-bias, so the recursion can be confined to a regime of small input bias without an exact-balancing step. Iterating amplifies the per-level logarithmic saving into a fixed exponent above $2/3$. The main text establishes the qualitative statement; the explicit value $\eta = 8 \cdot 10^{-4}$ is derived in Appendix A.

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2 Definitions and elementary composition facts

All Boolean functions are written in 0/1 notation. If V is a finite set, we identify an input $x \in \{0, 1\}^V$ with the set

$$X = \{v \in V : x_v = 1\}.$$

A function $f : \{0, 1\}^V \rightarrow \{0, 1\}$ is monotone if $X \subseteq Y$ implies $f(X) \leq f(Y)$.

For $p \in (0, 1)$, let $X \sim \mu_p$ denote the random subset of V in which each element is included independently with probability p . The p -biased total influence is

$$\text{Inf}_p(f) = \sum_{v \in V} \mathbb{P}_{X \sim \mu_p}[f(X) \neq f(X \triangle \{v\})].$$

The uniform total influence is the average sensitivity:

$$\text{AS}(f) = \text{Inf}_{1/2}(f).$$

For an input X , let $S(f, X)$ be the number of coordinates whose flip changes the value of f , and put

$$S(f) = \max_X S(f, X).$$

For monotone functions it is useful to separate the two sides:

$$S_0(f) = \max_{f(X)=0} |\{v \notin X : f(X \cup \{v\}) = 1\}|,$$

$$S_1(f) = \max_{f(X)=1} |\{v \in X : f(X \setminus \{v\}) = 0\}|.$$

Then

$$S(f) = \max\{S_0(f), S_1(f)\}.$$

The monotone dual of f is

$$f^\dagger(X) = 1 - f(V \setminus X).$$

Then

$$S_0(f^\dagger) = S_1(f), \quad S_1(f^\dagger) = S_0(f),$$

and

$$\text{Inf}_p(f^\dagger) = \text{Inf}_{1-p}(f).$$

If $f : \{0, 1\}^r \rightarrow \{0, 1\}$ and $g : \{0, 1\}^m \rightarrow \{0, 1\}$, their block composition $f \circ g$ is obtained by replacing each input bit of f by an independent copy of g . If

$$q = \mathbb{P}[g(X) = 1]$$

under the uniform measure, then

$$\text{Inf}(f \circ g) = \text{Inf}_q(f) \text{Inf}(g). \tag{1}$$

Indeed, a variable inside the i -th copy of g is pivotal exactly when it is pivotal for that copy of g , and the i -th coordinate of the outer function is pivotal at the vector of block outputs. The other block outputs are independent q -biased bits, and summing over all variables gives (1).

For monotone functions,

$$S_0(f \circ g) \leq S_0(f) S_0(g), \quad S_1(f \circ g) \leq S_1(f) S_1(g). \tag{2}$$

For example, at a zero-input of $f \circ g$, only upward-pivotal coordinates of the outer function can matter, and each corresponding block contributes at most $S_0(g)$ upward-pivotal inner variables. The proof on the one-side is identical.

3 A finite affine-cover lemma

We use the following finite-geometric fact.

Lemma 3 (Affine-cover bound). *Let $\mathbb{F}_Q^d$ be a d -dimensional affine space over the finite field $\mathbb{F}_Q$. Suppose affine hyperplanes*

$$H_1, \dots, H_m$$

cover $\mathbb{F}_Q^d$, and suppose the cover is irredundant: for every i , the hyperplane H_i contains a point not lying in any other H_j . Then

$$m \leq d(Q - 1) + 1.$$

Equivalently, in any affine-hyperplane cover of $\mathbb{F}_Q^d$, the number of essential hyperplanes is at most $d(Q - 1) + 1$.

Proof. Choose affine linear forms ℓ_i with

$$H_i = \{x : \ell_i(x) = 0\}.$$

The cover condition says that

$$P(x) := \prod_{i=1}^m \ell_i(x)$$

vanishes at every point of $\mathbb{F}_Q^d$. Work in the finite-function algebra

$$\mathcal{A}_d = \mathbb{F}_Q[y_1, \dots, y_d] / (y_1^Q - y_1, \dots, y_d^Q - y_d).$$

Every function on $\mathbb{F}_Q^d$ has a unique reduced representative in which each variable has degree at most $Q - 1$. The product above is therefore zero in $\mathcal{A}_d$.

Irredundancy means that deleting any one factor gives a nonzero function. Thus P is a minimal zero product of affine linear forms in $\mathcal{A}_d$. We use the following elementary form of the finite-field degree bound: a minimal zero product of affine linear forms in $\mathcal{A}_d$ has length at most $d(Q - 1) + 1$. This follows by reducing products in the basis of monomials

$$y_1^{a_1} \cdots y_d^{a_d}, \quad 0 \leq a_j \leq Q - 1.$$

Before the final multiplication the product is a nonzero reduced polynomial. A nonzero reduced polynomial has total degree at most $d(Q - 1)$, and multiplication by one further affine linear form is the only possible step at which the first zero product can occur. Hence the length of a minimal zero product is at most $d(Q - 1) + 1$. Applying this to P gives the desired bound. $\square$

4 The star-free algebraic DNF block

The main block is a DNF whose terms are polynomial graphs. The absence of the deterministic star is important: the zero-sensitivity is then governed by the affine-cover lemma instead of by the total number of variables.

Proposition 4 (Algebraic block). *Fix constants $0 < a < b < 1$ and $\Lambda \in (0, 1)$. There are constants $c_0, c_1, C_1 > 0$ and Q_0 , depending on a, b, Λ , such that for every prime power $Q \geq Q_0$ and every $p \in [a, b]$, there is a monotone Boolean function $R_{p,Q}$ satisfying*

$$c_0 \leq \mathbb{P}_{\mu_p}[R_{p,Q} = 1] \leq \Lambda,$$

$$\text{Inf}_p(R_{p,Q}) \geq c_1 Q,$$

and

$$S_1(R_{p,Q}) \leq Q, \quad S_0(R_{p,Q}) \leq C_1 \frac{Q^2}{\log Q}.$$

Proof. Let

$$D_a = \exp(a^{-1} - 1).$$

Choose an integer K so large that

$$D_a b^{K-1} \leq \frac{1}{4}, \quad b^{K-1} \leq \Lambda. \quad (3)$$

All constants below depend only on a, b .

Fix $p \in [a, b]$, and write

$$A = \log(1/p).$$

For sufficiently large Q , define

$$d = \left\lfloor \frac{(Q - K - 2)A}{\log Q} \right\rfloor,$$

and

$$L = \left\lfloor \frac{d \log Q}{A} + K \right\rfloor.$$

Then, uniformly in $p \in [a, b]$,

$$\frac{Q}{2} \leq L \leq Q \quad (4)$$

for all sufficiently large Q . Also

$$d \leq C \frac{Q}{\log Q}. \quad (5)$$

Choose a set $E \subseteq \mathbb{F}_Q$ with $|E| = L$. The variable set is

$$V = E \times \mathbb{F}_Q.$$

For every polynomial $P \in \mathbb{F}_Q[z]$ of degree less than d , define its graph over E by

$$\Gamma(P) = \{(t, P(t)) : t \in E\} \subseteq V.$$

Thus $|\Gamma(P)| = L$. Define

$$R_{p,Q}(X) = 1 \iff \Gamma(P) \subseteq X \text{ for some } P \in \mathbb{F}_Q[z], \deg P < d.$$

This is a monotone DNF whose terms all have size L .

First consider sensitivity. Since every term has size L ,

$$S_1(R_{p,Q}) \leq L \leq Q.$$

Indeed, if a term $\Gamma(P)$ is already present, deleting a coordinate outside that term does not destroy that witness.

Now let X be a zero-input and set

$$C = V \setminus X.$$

Since $R_{p,Q}(X) = 0$, every polynomial graph meets C . For each missing point $(t, y) \in C$, consider the affine hyperplane in the coefficient space $\mathbb{F}_Q^d$

$$H_{t,y} = \{P : \deg P < d, P(t) = y\}.$$

The family $\{H_{t,y} : (t,y) \in C\}$ covers all of $\mathbb{F}_Q^d$. Moreover, the point (t,y) is upward-pivotal for $R_{p,Q}$ at X precisely when $H_{t,y}$ is essential in this cover: there is a polynomial graph meeting C exactly at (t,y) . By Lemma 3, the number of such essential hyperplanes is at most

$$d(Q-1)+1.$$

Using (5),

$$S_0(R_{p,Q}) \leq C_1 \frac{Q^2}{\log Q}.$$

It remains to prove the density and influence bounds. Let $X \sim \mu_p$, and let

$$Z(X) = |\{P : \deg P < d, \Gamma(P) \subseteq X\}|$$

be the number of satisfied polynomial-graph terms. Since there are Q^d polynomials of degree less than d ,

$$\lambda := \mathbb{E}Z = Q^d p^L.$$

By the definition of L ,

$$L - \frac{d \log Q}{\log(1/p)} \in [K-1, K].$$

Therefore

$$a^K \leq \lambda \leq b^{K-1}. \tag{6}$$

We estimate the second factorial moment. For distinct polynomials P, R ,

$$|\Gamma(P) \cap \Gamma(R)| = |\{t \in E : P(t) = R(t)\}|.$$

For a nonzero polynomial S of degree less than d , define

$$z(S) = |\{t \in E : S(t) = 0\}|.$$

Then

$$\mathbb{E}[Z(Z-1)] = Q^d p^{2L} \sum_{\substack{S \neq 0 \\ \deg S < d}} p^{-z(S)}.$$

Put $\theta = p^{-1} - 1$. Since

$$p^{-z(S)} = (1 + \theta)^{z(S)} = \sum_{T \subseteq \{t \in E : S(t) = 0\}} \theta^{|T|},$$

we have

$$\sum_{\substack{S \neq 0 \\ \deg S < d}} p^{-z(S)} \leq \sum_{j=0}^{d-1} \binom{L}{j} \theta^j Q^{d-j}.$$

Indeed, after prescribing $j < d$ roots in E , there are at most Q^{d-j} degree- $< d$ polynomials satisfying those homogeneous linear conditions. Hence

$$\sum_{\substack{S \neq 0 \\ \deg S < d}} p^{-z(S)} \leq Q^d \left(1 + \frac{\theta}{Q}\right)^L \leq Q^d \exp(a^{-1} - 1) = D_a Q^d,$$

where we used $L \leq Q$ and $p \geq a$. Thus

$$\mathbb{E}[Z(Z-1)] \leq D_a \lambda^2.$$

By (3) and (6),

$$\mathbb{E}[Z(Z-1)] \leq \frac{1}{4} \lambda.$$

Since Z is integer-valued,

$$\mathbb{P}[Z = 1] \geq \mathbb{E}Z - \mathbb{E}[Z(Z-1)] \geq \frac{3}{4} \lambda \geq \frac{3}{4} a^K.$$

Consequently

$$\mathbb{P}_{\mu_p}[R_{p,Q} = 1] \geq \frac{3}{4} a^K.$$

The upper bound follows from Markov's inequality:

$$\mathbb{P}_{\mu_p}[R_{p,Q} = 1] = \mathbb{P}[Z \geq 1] \leq \mathbb{E}Z = \lambda \leq b^{K-1} \leq \Lambda.$$

Thus the density bounds hold with $c_0 = \frac{3}{4} a^K$.

Finally, on the event $Z = 1$, there is a unique satisfied graph $\Gamma(P)$. Deleting any of its L variables destroys the unique witness and cannot create a new one. Therefore

$$S(R_{p,Q}, X) \geq L$$

on this event. Since total influence is the expected sensitivity under μ_p ,

$$\text{Inf}_p(R_{p,Q}) \geq L \mathbb{P}[Z = 1] \geq \frac{Q}{2} \cdot \frac{3}{4} a^K \geq c_1 Q.$$

This proves the proposition. $\square$

The affine-cover bound $d(Q-1) + 1$ used in the proof above is sharp for the polynomial-graph construction: no further structural improvement of S_0 is possible within this family.

Lemma 5 (Tightness of the polynomial-graph zero-sensitivity bound). *Assume $L \geq d + 1$. Then there is a zero input X for $R_{p,Q}$ with*

$$S_0(R_{p,Q}, X) = d(Q-1) + 1.$$

Proof. Choose distinct points $t_1, \dots, t_d, t_* \in E$, and put

$$C = \{(t_i, y) : 1 \leq i \leq d, y \in \mathbb{F}_Q^\times\} \cup \{(t_*, 0)\}, \quad X := V \setminus C.$$

We first show $R_{p,Q}(X) = 0$: every polynomial P with $\deg P < d$ has $\Gamma(P)$ meeting C . Indeed, if $P(t_i) \neq 0$ for some i , then $(t_i, P(t_i)) \in C \cap \Gamma(P)$; otherwise P has d roots and $\deg P < d$, forcing $P \equiv 0$, and $(t_*, 0) \in C \cap \Gamma(P)$.

Now each point of C is upward-pivotal. For (t_i, y) with $y \neq 0$, let ℓ_i be the Lagrange basis polynomial of degree $d-1$ with $\ell_i(t_i) = 1$ and $\ell_i(t_j) = 0$ for $j \neq i$. The polynomial $y\ell_i$ has degree $< d$ and graph meeting C only at (t_i, y) (the value at t_* is $y\ell_i(t_*) \neq 0$ since $t_* \notin \{t_1, \dots, t_d\}$). For $(t_*, 0)$, the zero polynomial has graph meeting C only at $(t_*, 0)$. Hence $|C| = d(Q-1) + 1$ points are pivotal. $\square$

5 Biased cascade block

We combine two algebraic blocks under retuned biases to obtain a one-step gadget whose input and output distributions both lie in a fixed sub-interval $[p_-, \Lambda] \subset (0, 1)$ bounded away from 0 and 1. This closure on a positive interval replaces an exact-balancing step and keeps the dual inner block in a high-internal-bias regime, where the affine-cover sensitivity constant is small.

The proof of Proposition 4 actually gives a sharper estimate on the zero-side sensitivity. Tracking the explicit dependence of d on the input bias yields

$$S_0(R_{r,Q}) \leq (\log(1/r) + o(1)) \frac{Q^2}{\log Q}, \quad S_1(R_{r,Q}) \leq Q, \quad (7)$$

uniformly in r on any fixed compact subinterval of $(0, 1)$; the constant $\log(1/r)$ comes from $d \leq ((1 + o(1))Q \log(1/r)) / \log Q$ in the proof.

Fix absolute constants

$$0 < p_- < \Lambda < \frac{1}{2}, \quad r_0 := 1 - \Lambda, \quad r_+ := 1 - p_-,$$

so that $[r_0, r_+] \subset [1/2, 1)$. The values Λ, p_-, r_0, r_+ are calibrated by a Paley–Zygmund tuning of Proposition 4: choosing $\lambda = \Lambda$ and using the second-moment estimate of the proof, the output bias of $R_{r,Q}$ at internal bias r lies in $[p_-, \Lambda]$ whenever $r \in [r_0, r_+]$. The explicit values $\Lambda = 1/5, p_- > 0.127, r_0 = 4/5, r_+ < 0.873$ are derived in Appendix A.

Apply Proposition 4 on the interval $[r_0, r_+]$ with this upper bound Λ . There exist absolute constants $c_*, C_* > 0$ and a threshold Q_0 such that for every $r \in [r_0, r_+]$ and every prime power $Q \geq Q_0$,

$$p_- \leq \mathbb{P}_r[R_{r,Q} = 1] \leq \Lambda, \quad \text{Inf}_r(R_{r,Q}) \geq c_*Q, \quad (8)$$

together with (7), which on $[r_0, r_+]$ gives

$$S_0(R_{r,Q}) \leq (\log(1/r_0) + o(1)) \frac{Q^2}{\log Q} = C_* \frac{Q^2}{\log Q}. \quad (9)$$

Inner block. For input bias $p \in [p_-, \Lambda]$, define

$$B_{p,Q} := R_{1-p,Q}^\dagger.$$

Since $1 - p \in [r_0, r_+]$, the duality identities together with (8)–(9) give

$$\text{Inf}_p(B_{p,Q}) = \text{Inf}_{1-p}(R_{1-p,Q}) \geq c_*Q,$$

$$S_0(B_{p,Q}) = S_1(R_{1-p,Q}) \leq Q, \quad S_1(B_{p,Q}) = S_0(R_{1-p,Q}) \leq C_* \frac{Q^2}{\log Q},$$

and

$$q_p := \mathbb{P}_p[B_{p,Q} = 1] = 1 - \mathbb{P}_{1-p}[R_{1-p,Q} = 1] \in [r_0, r_+].$$

Outer block. Set $T_{q_p,Q} := R_{q_p,Q}$. Since $q_p \in [r_0, r_+]$, the same bounds apply:

$$\text{Inf}_{q_p}(T_{q_p,Q}) \geq c_*Q, \quad S_1(T_{q_p,Q}) \leq Q, \quad S_0(T_{q_p,Q}) \leq C_* \frac{Q^2}{\log Q}.$$

Cascade. Define

$$H_{p,Q} := T_{q_p,Q} \circ B_{p,Q}.$$

By the influence-composition identity (1),

$$\text{Inf}_p(H_{p,Q}) = \text{Inf}_{q_p}(T_{q_p,Q}) \text{Inf}_p(B_{p,Q}) \geq c_*^2 Q^2.$$

By the sensitivity-composition bound (2),

$$S_0(H_{p,Q}) \leq S_0(T_{q_p,Q}) S_0(B_{p,Q}) \leq C_* \frac{Q^3}{\log Q},$$

$$S_1(H_{p,Q}) \leq S_1(T_{q_p,Q}) S_1(B_{p,Q}) \leq C_* \frac{Q^3}{\log Q},$$

and hence

$$S(H_{p,Q}) \leq C_* \frac{Q^3}{\log Q}.$$

Finally, the cascade closes the recursion on $[p_-, \Lambda]$: under μ_p the inputs to $T_{q_p,Q}$ are i.i.d. q_p -biased, so by (8),

$$\mathbb{P}_p[H_{p,Q} = 1] = \mathbb{P}_{q_p}[T_{q_p,Q} = 1] \in [p_-, \Lambda].$$

Proposition 6 (Biased cascade block). *There exist absolute constants $0 < p_- < \Lambda < 1$, $c, C > 0$, and a prime-power threshold Q_0 such that for every prime power $Q \geq Q_0$ and every $p \in [p_-, \Lambda]$, there is a monotone Boolean function $H_{p,Q}$ with*

$$\mathbb{P}_p[H_{p,Q} = 1] \in [p_-, \Lambda], \quad \text{Inf}_p(H_{p,Q}) \geq cQ^2, \quad S(H_{p,Q}) \leq C \frac{Q^3}{\log Q}.$$

6 Recursive amplification

We freeze one sufficiently large Q and iterate the cascade. Each level retunes the outer block to the actual output bias of the previous level, and the cascade output remains in the fixed interval $[p_-, \Lambda]$ by Proposition 6.

By Proposition 6, for large Q ,

$$\frac{\log \text{Inf}_p(H_{p,Q})}{\log S(H_{p,Q})} \geq \frac{\log(cQ^2)}{\log(CQ^3/\log Q)} = \frac{2 \log Q + O(1)}{3 \log Q - \log \log Q + O(1)} = \frac{2}{3} + \frac{2 \log \log Q + O(1)}{9 \log Q + O(\log \log Q)},$$

which is strictly larger than $2/3$ for all sufficiently large Q and uniformly in $p \in [p_-, \Lambda]$. Choose one such prime power Q_* and a number $\rho > 2/3$ with

$$\text{Inf}_p(H_{p,Q_*}) \geq S(H_{p,Q_*})^\rho \quad \text{for every } p \in [p_-, \Lambda].$$

Pick $0 < \eta < \rho - 2/3$.

Let F_1 be any monotone Boolean function whose output bias under the uniform measure lies in $[p_-, \Lambda]$. An explicit example, valid for the values p_-, Λ of Appendix A, is the four-variable function

$$F_1(x_1, x_2, x_3, x_4) := x_1 \wedge x_2 \wedge (x_3 \vee x_4),$$

which has $\mathbb{P}[F_1 = 1] = 3/16 \in [p_-, \Lambda]$. For $m \geq 1$, set $p_m := \mathbb{P}[F_m = 1]$ and define recursively

$$F_{m+1} := H_{p_m, Q_*} \circ F_m.$$

By Proposition 6, $p_{m+1} \in [p_-, \Lambda]$, so the recursion is well-defined for every m .

Influence composition under the relevant intermediate bias gives

$$\text{AS}(F_{m+1}) = \text{Inf}(F_{m+1}) = \text{Inf}_{p_m}(H_{p_m, Q_*}) \text{Inf}(F_m) \geq cQ_*^2 \cdot \text{AS}(F_m).$$

The sensitivity-composition bound (2) gives

$$\text{S}(F_{m+1}) \leq \text{S}(H_{p_m, Q_*}) \text{S}(F_m) \leq C \frac{Q_*^3}{\log Q_*} \cdot \text{S}(F_m).$$

Iterating,

$$\text{AS}(F_m) \geq (cQ_*^2)^{m-1} \text{AS}(F_1), \quad \text{S}(F_m) \leq \left(C \frac{Q_*^3}{\log Q_*}\right)^{m-1} \text{S}(F_1).$$

Therefore

$$\text{AS}(F_m) \geq \text{S}(F_m)^{2/3+\eta}$$

for all sufficiently large m , and after reindexing the sequence, for every $m \geq 1$. Since $cQ_*^2 > 1$ we have $\text{AS}(F_m) \rightarrow \infty$, and hence $\text{S}(F_m) \rightarrow \infty$.

This completes the proof of Theorem 2.

A An explicit value of η

The main text proves only the existence of an absolute constant $\eta > 0$. In this appendix we sharpen the estimates of Propositions 4 and 6 and pin down a concrete value

$$\rho > 0.66757, \quad \eta = 8 \cdot 10^{-4},$$

so that

$$\text{AS}(F_m) \geq \text{S}(F_m)^{2/3+8 \cdot 10^{-4}}.$$

The argument has the same structure as in the main text; only the constants are tightened.

A.1. The Poisson tuning and the closure interval

Fix

$$\Lambda = \frac{1}{5}, \quad r_0 = \frac{4}{5}, \quad D_0 = \exp(r_0^{-1} - 1) = e^{1/4} \approx 1.28403.$$

In the proof of Proposition 4, applied on $[r_0, r_+] \subset [r_0, 1)$, choose K so that the density parameter $\lambda = Q^d r^L$ satisfies

$$r \Lambda \leq \lambda \leq \Lambda,$$

which gives $\lambda \in [r_0 \Lambda, \Lambda] = [4/25, 1/5]$ for $r \in [r_0, r_+]$. The second-moment estimate (6) reads $\mathbb{E}[Z(Z-1)] \leq D_0 \lambda^2$, so Paley–Zygmund gives

$$\mathbb{P}_r[R_{r,Q} = 1] \geq \frac{(\mathbb{E}Z)^2}{\mathbb{E}Z^2} \geq \frac{\lambda}{1 + D_0 \lambda} \geq \frac{r_0 \Lambda}{1 + D_0 \Lambda} =: p_- > 0.127.$$

Set $r_+ := 1 - p_- < 0.873$. The output bias of $R_{r,Q}$ at any internal bias $r \in [r_0, r_+]$ therefore lies in $[p_-, \Lambda]$, and the dual identity $\mathbb{P}_p[B_{p,Q} = 1] = 1 - \mathbb{P}_{1-p}[R_{1-p,Q} = 1]$ places q_p back into $[r_0, r_+]$. This is the closure of the cascade asserted in Section 5.

A.2. A sharper influence bound for the algebraic block

The proof of Proposition 4 actually yields the second-moment estimate

$$\text{Inf}_r(R_{r,Q}) \geq \frac{L}{r}(\lambda - D_0\lambda^2 - o(1)).$$

The first moment of pairs (P, v) with $v \in \Gamma(P)$ and $\Gamma(P) \setminus \{v\} \subseteq X$, $v \notin X$ is exactly $L\lambda/r$; the overcount from variables pivotal at two distinct witnesses is at most $(L/r)D_0\lambda^2$. With $\lambda \geq r_0\Lambda$, $\lambda \leq \Lambda$, $L = (1 + o(1))Q$, and $r \leq r_+$,

$$\text{Inf}_r(R_{r,Q}) \geq (1 - o(1))Q \cdot \frac{r_0\Lambda}{r_+}(1 - D_0\Lambda) \geq 0.135Q.$$

Explicit evaluation gives $(r_0\Lambda/r_+)(1 - D_0\Lambda) > 0.136$.

A.3. The cascade constant

The cascade applies the bound of Section A.2 twice—once to the inner block at internal bias $1 - p \in [r_0, r_+]$ and once to the outer block at internal bias $q_p \in [r_0, r_+]$. Influence composition therefore yields

$$\text{Inf}_p(H_{p,Q}) \geq (0.135)^2 Q^2 (1 - o(1)) = 0.018225 Q^2 (1 - o(1)).$$

Hence we may use

$$c = 0.018225$$

in Proposition 6.

A.4. The sensitivity constant

Both $B_{p,Q}$ and $T_{q_p,Q}$ are evaluated at internal bias in $[r_0, r_+]$, so (7) gives

$$S_0(R_{r,Q}) \leq (\log(1/r) + o(1)) \frac{Q^2}{\log Q} \leq (\log \frac{5}{4} + o(1)) \frac{Q^2}{\log Q}$$

on each side, and consequently

$$S(H_{p,Q}) \leq (\log \frac{5}{4} + o(1)) \frac{Q^3}{\log Q}.$$

Hence we may take

$$C = \log(5/4) \approx 0.22314.$$

For comparison, an exact-balancing step (avoided by the cascade) would multiply this by $2 \log 2 / \log(5/4) \approx 6.21$; avoiding the balancing gadget and pushing the dual bias to $\geq 4/5$ is the dominant source of the improvement.

A.5. Optimization

Combining the three estimates,

$$\rho(Q) \geq \frac{\log(cQ^2)}{\log(CQ^3/\log Q)} + o(1) = \frac{2 \log Q + \log c}{3 \log Q - \log \log Q + \log C} + o(1).$$

Writing $x = \log Q$, the function

$$\rho(x) = \frac{2x + \log c}{3x - \log x + \log C}$$

with $c \approx 0.018225$ and $C \approx 0.22314$ is maximized near $x \approx 245$. Taking $Q = 2^{354}$ (so that $\log Q \approx 245.37$) gives

$$\rho(Q) \approx 0.66757, \quad \rho(Q) - \frac{2}{3} \approx 8.9 \cdot 10^{-4}.$$

Rounding down to absorb floor errors and the $o(1)$ terms, we obtain the rigorous values

$$\rho > 0.66757, \quad \eta = 8 \cdot 10^{-4}.$$

A.6. Structural bottleneck

This is the best improvement that the polynomial-graph and affine-cover method gives without introducing a fundamentally new seed construction. By Lemma 5, the bound $S_0(R_{r,Q}) \leq d(Q-1) + 1$ is sharp for this family of DNFs, so the asymmetry $S_0 \sim Q^2/\log Q$ versus $S_1 \sim Q$ and $\text{Inf} \sim Q$ cannot be tightened by sharpening the cover lemma. After dual pairing, the cascade is forced into the shape $\text{Inf} \sim Q^2$, $S \sim Q^3/\log Q$, and the exponent extraction has the form

$$\frac{2 \log Q + O(1)}{3 \log Q - \log \log Q + O(1)} = \frac{2}{3} + O\left(\frac{\log \log Q}{\log Q}\right).$$

Freezing one finite Q converts this logarithmic saving into a small fixed η . A genuinely larger improvement would require a new term system simultaneously achieving $\text{Inf} \gg L^{1-o(1)}$, $S_1 \lesssim L$, and $S_0 \ll L^2/\log L$.

References

- [FHH⁺14] Y. Filmus, H. Hatami, S. Heilman, E. Mossel, R. O’Donnell, S. Sachdeva, A. Wan, and K. Wimmer, *Real analysis in computer science: A collection of open problems*, Simons Institute for the Theory of Computing, 2014.
- [OS07] R. O’Donnell and R. A. Servedio, Learning monotone decision trees in polynomial time, *SIAM J. Comput.* 37(3):827–844, 2007.
- [ScTa13] D. Scheder and L.-Y. Tan, On the average sensitivity and density of k -CNF formulas, in *Approximation, Randomization, and Combinatorial Optimization (APPROX/RANDOM 2013)*, Lecture Notes in Computer Science, vol. 8096, Springer, Berlin, Heidelberg, 2013, pp. 683–698.